MENDSPEECH REVISED LEARNING SYSTEM

MendSpeech Project Blueprint

Selective semantic speech restoration under real time constraints, with cascaded and direct audio repair baselines.

> Core principle: preserve trustworthy original speech, spend compute only where uncertainty justifies it, and never hide architectural limitations.

1. Product Behavior

- Accept microphone audio or an uploaded recording.
- Optionally generate controlled damage through SpeechDamageBench.
- Transcribe incrementally with cache aware FastConformer inference.
- Align confidence and uncertainty to time.
- Classify intervals as Preserve, Inspect, Repair, or Abstain.
- For MendSpeech V1, reconstruct only repair intervals with speaker conditioned TTS, then apply boundary matching before stitching.
- For the research comparison, run the same damaged spans through a pretrained direct latent or codec audio inpainting baseline.
- Show exactly which milliseconds were preserved, reconstructed, or left unrepaired.

2. Two Repair Architectures

MendSpeech V1: Cascaded

Streaming ASR to calibrated uncertainty to repair policy to speaker conditioned TTS to duration alignment to acoustic boundary matching to waveform stitching.

Strength: deterministic semantic control and a practical real time systems path.

Limitation: text discards pitch, emotion, breathing, coarticulation, and other acoustic context.

Direct Audio Baseline

Damaged audio to latent or codec representation to masked or conditioned reconstruction to restored audio.

Strength: can preserve more acoustic context without a text bottleneck.

Limitation: may require more compute, be harder to control semantically, or be less suitable for strict streaming constraints.

3. Boundary Matching Layer

- Select a repair window with context padding.
- Match generated duration to the target interval.
- Compare short time energy around both boundaries.
- Match local loudness before mixing.
- Estimate simple spectral mismatch and room tone difference.
- Compare linear and equal power crossfades.
- Record seam diagnostics so subjective listening is not the only evidence.

4. SpeechDamageBench

SpeechDamageBench is a standalone package, not a private utility inside MendSpeech.

Damage family	Controlled variables	Purpose
Additive noise	SNR, noise type, seed	Test masking robustness.

Clipping	Threshold, severity	Test lost peaks and saturation.
Bandwidth limits	Sample rate, filter settings	Simulate narrow channels.
Dropouts	Span length, frequency, seed	Simulate missing speech and packet loss.
Reverberation	Impulse response or severity	Test temporal smearing.

> Every generated sample records corruption name, severity, seed, clean source id, parameter values, and package version.

5. Research Console Metrics

Metric	Why it matters
WER and CER	Recognition correctness.
Latency percentiles	Responsiveness and tail behavior.
Real time factor	Whether processing keeps up with speech.
GPU memory	Deployment pressure.
Repair percentage	How much audio was regenerated.
Original retained percentage	Preservation directly.
Speaker similarity proxy	One imperfect signal for identity consistency.
Calibration	Whether confidence supports policy decisions.
Boundary energy discontinuity	One signal for stitching quality.
Abstention outcomes	Whether the system avoids inventing unrecoverable content.

6. Required Ablations

- Fixed low lookahead versus fixed high lookahead versus adaptive context.
- Raw confidence versus calibrated confidence.
- Preserve, Balanced, and Rescue thresholds.
- Selective repair versus full utterance resynthesis.
- Naive stitching versus boundary matched stitching.
- Base ASR versus robustness adapted ASR.
- Cascaded V1 versus direct latent or codec audio repair.
- Short versus long missing spans.
- Clean speech regression to ensure already good audio is not damaged.

7. Repository Target

```
mendspeech/
  src/audio/
  src/asr/
  src/streaming/
  src/controller/
  src/tts/
  src/repair/
  src/baselines/
  src/metrics/
  src/bench/
  infra/
  app/
  experiments/
  results/
  reports/

speechdamagebench/
  speechdamagebench/
  tests/
  presets/
  pyproject.toml
  README.md
```

8. Definition of Done

- A new user can reproduce one benchmark with documented commands.
- The demo makes preserved and reconstructed spans visually obvious.
- Boundary matching is measured, not described only by listening.
- The report contains at least one result that surprised you and one limitation that materially constrains the claim.
- The system abstains when content is too uncertain.
- The direct audio baseline is treated fairly and the report says where it is stronger.
- You can explain every major model and system component without relying on framework names as the explanation.